

Methods

All samples were cut into rectangular slabs of 2 cm x 3 cm. These were subsequently placed on a coverslip and ground down to a thickness of 30 μm . The massive sulphide samples were cut into 3 cm x 2 cm x 1 cm cuboids. These were placed in epoxy resin and were subsequently polished. All LA-ICPMS analysis were performed at the Department of Geosciences (University of Mainz) with an ESI NWR193 ArF 193nm excimer laser system equipped with a TwoVol² ablation cell coupled to an Agilent 7500ce quadrupole ICPMS.

Analyses of coherent basalt and volcaniclastics

Bulk rock data

Bulk rock data was acquired combining X-Ray-Fluorescence (XRF) analysis and LA-ICPMS analysis on whole-rock glass shards. Major and some trace element concentrations were determined using XRF analysis applying a Philips MagiXPRO spectrometer with a rhenium X-ray tube (Department of Geosciences, University of Mainz). LA-ICPMS trace element analysis was performed on glass beads using an iridium strip heater according to Stoll et al. (2008). During fusion and quenching, the reaction chamber was flooded with argon gas to prevent oxidation. Because of volatilization during melting, the glass beads are to some extent depleted in volatile elements. Due to the high SiO_2 content of samples 16B10 and 16B11, MgO was added to lower their respective melting points (Nehring et al. 2008). The glass beads were subsequently analysed using LA-ICPMS. The ArF LA system was operated at a pulse repetition rate of 10 Hz and an energy density of c. 3 J cm^{-2} . Randomly distributed across the surface of each sample, six to eight spots with a diameter of 100 μm were analysed performing background measurement for 15 s, ablation times of 30 s and wash-out times of 20 s. Ablation was carried out under a He atmosphere and the sample gas was mixed with Ar before entering the plasma. For calibration we used the

multi-element synthetic glass NIST SRM 610 applying the preferred values reported in the GeoReM database (<http://georem.mpch-mainz.gwdg.de>) (Jochum et al. 2005; 2011) as the “true” concentrations to calculate the element concentrations of the samples. Additionally, we analysed NIST SRM 612, USGS BCR-2G, USGS GSD-1G and MPI-DING StHs6/80-G as quality control materials (QCMs) to monitor precision and accuracy of the measurements as well as calibration strategy. Signals were monitored in time-resolved mode and processed using GLITTER 4.4.1 (Griffin et al. 2008) and ^{29}Si as internal standard applying the SiO_2 contents previously determined by XRF from aliquots of the same powders and the preferred values reported in the GeoReM data base for all QCMs. Resulting element concentrations for the QCMs together with reference values are given in Online Resource 3.

Zircon analysis

Volcaniclastic sample 16B10 was crushed to grain sizes below 500 μm and elutriated to get rid of small particles ($<63 \mu\text{m}$). Zircon crystals were handpicked under a binocular with the help of UV-light. Zircon grains were mounted in epoxy resin, polished to their centres, carbon coated and imaged by cathodoluminescence (CL) and back-scattered electron (BSE) using a scanning electron microscope (SEM; JEOL JSM-6610LV with a EDS analytical system) at the University of Belgrade. Ion intensities for the following isotopes were measured by LA-ICPMS applying a pulse repetition rate of 10 Hz and an energy density of about 3 J cm^{-2} : ^{202}Hg , $^{204}(\text{Pb}, \text{Hg})$, ^{206}Pb , ^{207}Pb , ^{208}Pb , ^{232}Th and ^{238}U . Background measurements of 15 s were followed by 30 s of ablation and 20 s of wash out time. Single spot measurements with a diameter of 30 μm were arranged along transects through the zircon crystals with emphasis on sampling rim and core regions (all analysis spot locations are marked in Online Resource 4). Ablation locations were selected based on CL images. For calibration of the LA-ICP-MS measurements, Plesovice zircon was used, while 91500

and GJ1 zircons were used as QCM to monitor accuracy and precision of the measurements. Data reduction was done, using the LaDating@Zircon (Hou 2015) excel spreadsheet in combination with the Isoplot (4.15) add-in for excel (Ludwig 2012). Obtained data for the zircon 91500 are in agreement with published ages but display a broad range from 1020 to 1100 Ma. The calculated concordia age is 1050.8 ± 6.0 Ma, while the age in the literature is 1065 Ma. This represents a mean deviation of 1.3 % from the published age. Since the data scatter of the analysed zircons are larger than this, we regard this as negligible.

Analyses of massive sulphides

Major and trace element analysis

Major element compositions of sphalerites were investigated using a JEOL JXA-8200 electron microprobe with a Wavelength Dispersive Spectrometry (WDS) analytical system at the Johannes Gutenberg-University, Mainz. Analyses were performed at 20 kV accelerating voltage, 20 nA current and an electron-beam diameter of 5 μm . As reference materials aAg₂Te, aCdS, aCuFeS₂, aHgS, aInSb, aMnTi, aPbS, aWOLLA, aZnS, bBiSe, bCOBALTIN, bPENTLAND, cMo and cRe were used. The obtained mean content of 65.40 wt.% for Zn in sphalerite was used as internal standard for LA-ICPMS analysis.

Trace element concentrations in sphalerite have been determined utilizing the LA-ICPMS set-up mentioned above. The laser was set to a pulse repetition rate of 10 Hz and the energy density was kept at 3.0 J/cm² using a spot size of 40 μm . Background signals were monitored for 15 s and the ablation signal for 35 s. To ensure low background signals throughout the experiment, the ablation cell was subsequently washed out with a pure gas for 20 s after each spot measurement. Reference materials were analysed for calibration and quality control prior and subsequent to each analysis

sequence and after every 30th analysis during sequences. The following isotopes were monitored: ^{33}S , ^{55}Mn , ^{57}Fe , ^{59}Co , ^{60}Ni , ^{65}Cu , ^{66}Zn , ^{71}Ga , ^{74}Ge , ^{75}As , ^{82}Se , ^{107}Ag , ^{111}Cd , ^{115}In , ^{118}Sn , ^{121}Sb , ^{197}Au , ^{202}Hg , ^{205}Tl and ^{208}Pb , and ^{209}Bi with an integration time of 10 ms for each mass. Data reduction was done using GLITTER 4.4.1 (Griffin et al. 2008) with Zn as internal standard. For most elements the sintered MUL-ZnS-1 sphalerite reference material (Onuk et al. 2017) was used for calibration while the pressed powder pellet USGS MASS-1 (Wilson et al. 2002) and the silicate glass NIST SRM 610 was used as QCM. However, there are no concentration data available for some elements of interest for this study in MUL-ZnS-1. Therefore, we used USGS MASS-1 for calibration of Hg and Bi and NIST SRM 610 for Au, while the other reference material served as QCM. For the reference materials the values given in the GeoReM database (<http://georem.mpch-mainz.gwdg.de/>; Jochum et al. 2005) were applied. Due to small grain sizes, monitored signals are irregular and display mixing with other sulphide phases. The time-resolved analysis allows for selection of undisturbed signal sections. Previous microprobe measurements didn't display any variation in major element composition for different sphalerite grains. Therefore, the median for each trace element of each sulphide sample was calculated and is expected to be the most representative. In total 37 trace element spot analyses were performed in sphalerite.

Sulphur isotopes – pyrite and barite

Sulphur isotope measurements were performed at the Scientific and Technological Center at the University of Barcelona. Due to the heterogeneity and small grain sizes of the sulphides, simple grinding and subsequent hand-picking under a binocular microscope were not possible. Instead, an ESI New Wave micromill was applied using a drill head with a diameter of 300 μm . Since this is the smallest available size, it was only possible to retrieve clean samples from pyrite crystals (Py2) larger than 500 μm and large enough barite veins. A drill depth of 200 μm in pyrite was

assumed to be reasonable, to avoid a possible drill through and hence potential contamination. Prior to milling, crystals of interest were counter-checked with reflected light pictures and discarded if they displayed inclusions or cracks. After each drill run, the drill head and the sample were thoroughly cleaned with ethanol to evade possible cross contamination. For the followed sulphur isotope analysis, a minimum of 1 mg sample was necessary. This was not possible to realize for every single sample. Pyrite powders from micro-drilling were therefore mixed from samples 16B06, 16B07 and 16B09_2. Additionally, barite veins from samples 16B06 and 16B07 were sampled individually through micro-drilling. Barite in 16B09_2 was impossible to sample due to small grain sizes. Sample 16B12, a pure barite vein was crushed with a hammer and crystals were hand-picked avoiding zones of alteration and disseminated sulphides. All barite samples originate from the same horizon within the massive sulphide mineralization. Samples were weighed into individual tin capsules. The capsules were sealed and burned in Elemental Analyser Carlo Erba EA 1108, where all sulphur oxidized and sulphate was reduced to SO₂. The produced SO₂ was subsequently analysed for sulphur isotope composition in an isotope ratio mass spectrometer (IRMS) Delta Plus XP (ThermoFisher). Used reference materials were: IAEA SO-5 ($\delta^{34}\text{S} = +0.5 \text{ ‰}_{\text{VCDT}}$), IAEA SO-6 ($\delta^{34}\text{S} = -34.1 \text{ ‰}_{\text{VCDT}}$) and NBS127 ($\delta^{34}\text{S} = +20.3 \text{ ‰}_{\text{VCDT}}$). The resulting $\delta^{34}\text{S}$ value is an expression of the $^{34}\text{S}/^{32}\text{S}$ ratio relative to V-CDT (Vienna Canyon Diablo Troilite). Delta values are given in permil.

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